

Applied Deep Learning

Image Classification, Convolution, Pooling, and Validation

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Assignment 7

Introduction

This report will answer 10 short questions based on assigned readings about image classification, convolutional neural network, pooling and model validation and error analysis.

Each question will explain the definitions and examples in simple language.

Question 1: What does image classification mean? Explain with one example.

Image classification is the task which assigns a label or category to a complete image. The model looks at the pixel values of an image and decide which class it belongs to from the pixel images options and select it. The model does not tell where is the image from, it just simply answers the questions “what is the image?”

It works as image is fed into the model as pixel and the model processes the image with multiple layers, then final layer outputs a probability for each class and then make a prediction. For example: If the input is the image of shoe then model will processes and give an output of Sneaker.

Question 2: Why is using a Dense Network not always a suitable choice for images?

A Dense Network connects every input neuron to every output neuron. For images this create a serious problem.

Issue 1: Too many parameters

- A small image of $28 \times 28 = 784$ pixels as inputs
- A medium size image $224 \times 224 = 150,528$ pixels in inputs
- One layer of 512 neurons= $150,528 \times 512 = 77$ millions weights– even just in 1 layer
- This is very slow to train and make their memory.

Issue 2: Losses structure

- Dense networks flatten the image into 1D list
- Then it destroyed the 2D structure and model will no longer knows which pixel image are close
- For example: A dog's eye being looks so different on the top-left and bottom-right

Issue 3: Would not detect the pattern

- Dense networks treat all pixels in same way(Equally). They cannot focus on small size pictures like edges or textures

Question 3: In a CNN, what is convolution, and what does a filter or kernel do?

Convolution is the mathematical operation of sliding a small set of numbers across the image and compute that dot into that position. For example: A filter is placed on the top left corner of the image and each filter value is multiplied by the pixel values of numbers and then all those products are added together to give a final output. The filter slides one step to the right and this pattern repeats until the filter has covered the complete image. Next, about filter, different filters detect different features like curves, edges, and corners. In early stages, it shows simple layers but in deep layers it's complex in features and network learns from the best values during training.

Question 4: What is a feature map in a convolutional neural network?

A feature map is also called an activation map is the output grid which is produced after one filter has been moved across the entire image. It is produced by one filter across the whole image and at each position, it produces 1 number. These all numbers together form a 2D grid, so this is the feature map. Feature map represents in different ways. If you have 32 filters, you will get 32 feature maps, each one detected a different feature.

Question 5: Why is weight sharing used in CNNs? Explain.

Weight sharing means the same filter is used for every position across the entire image. The filter does not change when it's moved. It always uses as the same number. It is powerful because it dramatically reduces the parameters means instead of unique weight for every pixel position, you would only need to learn weight of a small filter. It also strong for translation invariance like if a cat ear appears in the top-left or bottom-right the same filter detect it. It also prevents overfitting because by this way there is less chances of memorising the noise.

Question 6: How does the ReLU activation function work? Write the ReLU output for the following values: [-3, 0, 2, 5]

ReLU stands for Rectified linear unit. It is an activation function which is applied after convolution to introduce a non-linear activity into the network. The formula is very simple as if the value is positive then keep it as is. If the value is negative then replace it with zero. Formula: $\text{ReLU}(x) = \max(0, x)$

Input: [-3, 0, 2, 5]

If $x = -3$,

$\text{ReLU}(-3) = \max(0, -3) = 0$

If $x = 0$,

$\text{ReLU}(0) = \max(0, 0) = 0$

If $x = 2$,

$\text{ReLU}(2) = \max(0, 2) = 2$

If $x = 5$

$\text{ReLU}(5) = \max(0, 5) = 5$

So the output: [0, 0, 2, 5]

Without any activation function, stacking layers is mathematically the same as one layer so the neural network cannot learn from complex patterns. ReLU added non-linearity so the network can learn from curves, shapes and from complex patterns. Its just a simple comparison.

Question 7: What is max pooling, and why is it used in CNNs?

Max pooling is a down sampling operation. It divides the feature map into small regions and keeps the only maximum values from each region. For example: lets take a 2x2 window front eh feature map and pick the largest number from those 4 values and then move the window across the entire feature map, then the result would be a small feature map. Like a 2x2 window values

1, 3, 2, 0 ———> Max pooling output is 3

4, 6, 1, 3 ———> Max pooling output is 6

Question 8: In the Fashion-MNIST code, why are images reshaped from 28×28 to $28 \times 28 \times 1$?

The CNNs process images as 3D arrays of numbers. The 3 dimensional are height x width x channels. Channels represents the depth of the colours of the images. Grayscale have 1 channel and colour (RGB) image have 3 channels which are Red, Green, blue. Fashion-MNIST images comes as 28×28 in 2D arrays. Keras conv2D layers requires 3D input in height, width nd channel so without reshaping the model throws a dimensional error, so we added the 1 to explicitly tell the model that it is 1 channel. Then it comes $28 \times 28 \times 1$ instead if 28×28 .

Question 9: What is the difference between training data, validation data, and test data?

When building a machine learning model, data is split into three separate sets and each sets with a different purpose.

Training Data: The model learns from training data. Its weights are updated based on errors here. It has typically 70 to 80% of total dataset.

Validation Data: Validation data used during training to monitor performance. It also helps to tune the learning rate and layers. Model does not learn from validation data. This is typically 10 to 15%.

Test Data: Test data is only used once at the very end point. It gives the final unbiased accuracy and the model has never seen this data before, it typically 10 to 15%.

Every level of data has its own importance on different stages.

Question 10: What is overfitting? If training accuracy is high but validation accuracy is low, what does this indicate?

Overfitting happens when the model learns from the training data too well and it includes its noise and random patterns rather than learning front eh general rules.

Overfitting happens when high training accuracy adds with low validation accuracy.

The model has memorised the training data rather than learning from patterns so it performs perfectly on data it has seen, but fails on new unseen dataset. It comes when training accuracy keeps rising but validation accuracy stops improving or drops and large gap between training and validation accuracy developed. To fix this issue, we should add more training data and use dropout layers and simplify the model so it learns better and use early stop to keeps the accuracy.